with boron or some boron compound, and the tube is used in the proportional region, so that it will count alpha particles only. As the neutrons strike the boron layer, they produce there the following nuclear reaction:

$$_5B^{10} + _0n^1 \rightarrow _3Li^7 + _2He^4$$

This reaction produces fast helium nuclei, i.e., artificial alpha particles—one alpha particle per neutron. Every neutron evoking a nuclear reaction causes the counter to react with an impulse, in which the lithium nucleus will also participate. Not every neutron, by any means, striking the counter causes a nuclear reaction; many of them traverse the tube without any effect whatsoever. Nevertheless, the counter registers a number of neutrons proportional to the actual total number of neutrons. The constant factor of proportionality being as yet unknown.

Another frequently employed method consists in placing a *tracer* at the point where neutrons are suspected to be present. A tracer is a substance—for example, a piece of silver foil—made artificially radioactive by a neutron-induced nuclear reaction. The following two reactions will then take place in the silver, in this order:

(1) $_{47}Ag^{107} + _0n^1 \rightarrow _{47}Ag^{108}$

(2) $_{47}Ag^{108} \rightarrow _{48}Cd^{108} + _{-1}e^0$

In other words, the silver isotope of mass number 107 first changes into another silver isotope, which has the mass number 108. The latter is unstable, has a half-life of 22 seconds, and emits an electron and thus changes into a nuclear isobar, the cadmium atom $_{48}Cd^{108}$. The $_{47}Ag^{108}$ atom must be unstable, because its nucleus has 61 neutrons and 47 protons, in other words, it is a ‘doubly odd’ nucleus.

Since, as we have already seen, slow neutrons are usually more prone to capture by a nucleus than are fast neutrons, a boron counter will register a greater number of slow neutrons than fast ones. If a counter of this type is brought into the vicinity of a source of fast neutrons, a loudspeaker can be used, which is capable of making the individual impulses audible at a certain average rate, say, one per second. As we already know, neutrons can be slowed down by being made to pass through a hydrogen-containing substance, such as paraffin. If the counter is surrounded by paraffin, the impulses will multiply very considerably and a crackling sound will be audible. Thus, contrary to the naive assumption that the application of